

About

Agent2Agent (A2A) is an open protocol enabling communication and interoperability between opaque agentic applications.

[a2a-protocol.org/](#)

#agents #linux-foundation #a2a #generative-ai #a2a-protocol #a2a-mcp #a2a-server

 Readme

 Apache-2.0 license

 Code of conduct

 Contributing

 Security policy

 Activity

 Custom properties

 24.5k stars

 233 watching

 2.5k forks

Report repository

Releases 11

 v1.0.1 Latest
on May 28

[+ 10 releases](#)

Contributors 159



[+ 145 contributors](#)

Languages



 .devcontainer	docs(spec): Large refactor of sp...	7 months ago
 .gemini	ci: Add custom config for gemi...	9 months ago
 .github	ci: trigger docs build on change...	last week
 .mkdocs	docs: update a2a homepage ba...	last week
 .vscode	docs(spec): Large refactor of sp...	7 months ago
 adrs	docs: Added adr template and ...	5 months ago
 docs	docs: update a2a homepage ba...	last week
 scripts	ci: Minor updates for workflows...	last month
 specification	docs: add multi-tenancy guide ...	last month
 .editorconfig	chore: Add .editorconfig to set ...	11 months ago
 .git-blame-ignore-revs	chore: add .git-blame-ignore-re...	last year
 .gitattributes	docs(spec): Large refactor of sp...	7 months ago
 .gitignore	ci(linter): Add API Linter job to ...	4 months ago
 .gitvote.yml	Update .gitvote.yml (#1297)	6 months ago
 .prettierrc	docs: Homepage Style Updates ...	last year
 .ruff.toml	ci: Update Super Linter to only ...	9 months ago
 CHANGELOG.md	chore(main): release 1.0.1 (#1749)	last month
 CODE_OF_CONDUCT.md	docs: Fix link redirect issues (#1...	4 months ago
 CONTRIBUTING.md	ci: Update guidelines for commi...	last month
 GOVERNANCE.md	docs(GOVERNANCE.md): How t...	2 months ago
 LICENSE	Initial commit	last year
 MAINTAINERS.md	chore: Update MAINTAINERS.m...	9 months ago
 README.md	docs: restructure homepage inf...	2 weeks ago
 SECURITY.md	docs(spec): Large refactor of sp...	7 months ago
 lychee.toml	docs: Fix link redirect issues (#1...	4 months ago

Agent2Agent (A2A) Protocol

pypi v1.1.0 License Apache 2.0  Code Wiki

►  Language



Agent2Agent (A2A) Protocol

An open protocol enabling communication and interoperability between opaque agentic applications.

The Agent2Agent (A2A) protocol addresses a critical challenge in the AI landscape: enabling gen AI agents, built on diverse frameworks by different companies running on separate servers, to communicate and collaborate effectively - as agents, not just as tools. A2A aims to provide a common language for agents, fostering a more interconnected, powerful, and innovative AI ecosystem.

With A2A, agents can:

- Discover each other's capabilities.
- Negotiate interaction modalities (text, forms, media).
- Securely collaborate on long-running tasks.
- Operate without exposing their internal state, memory, or tools.



Join this short course on [A2A: The Agent2Agent Protocol](#), built in partnership with Google Cloud and IBM Research, and taught by [Holt Skinner](#), [Ivan Nardini](#), and [Sandi Besen](#).

What you'll learn:

- **Make agents A2A-compliant:** Expose agents built with frameworks like Google ADK, LangGraph, or BeeAI as A2A servers.
- **Connect agents:** Create A2A clients from scratch or using integrations to connect to A2A-compliant agents.
- **Orchestrate workflows:** Build sequential and hierarchical workflows of A2A-compliant agents.
- **Multi-agent systems:** Build a healthcare multi-agent system using different frameworks and see how A2A enables collaboration.
- **A2A and MCP:** Learn how A2A complements MCP by enabling agents to collaborate with each other.

Why A2A?

As AI agents become more prevalent, their ability to interoperate is crucial for building complex, multi-functional applications. A2A aims to:

- **Break Down Silos:** Connect agents across different ecosystems.
- **Enable Complex Collaboration:** Allow specialized agents to work together on tasks that a single agent cannot handle alone.
- **Promote Open Standards:** Foster a community-driven approach to agent communication, encouraging innovation and broad adoption.

- **Preserve Opacity:** Allow agents to collaborate without needing to share internal memory, proprietary logic, or specific tool implementations, enhancing security and protecting intellectual property.

Key Features

- **Standardized Communication:** JSON-RPC 2.0 over HTTP(S).
- **Agent Discovery:** Via "Agent Cards" detailing capabilities and connection info.
- **Flexible Interaction:** Supports synchronous request/response, streaming (SSE), and asynchronous push notifications.
- **Rich Data Exchange:** Handles text, files, and structured JSON data.
- **Enterprise-Ready:** Designed with security, authentication, and observability in mind.

Getting Started

-  **Explore the Documentation:** Visit the [Agent2Agent Protocol Documentation Site](#) for a complete overview, the full protocol specification, tutorials, and guides.
-  **View the Specification:** [A2A Protocol Specification](#)
- Use the SDKs:
 -  [A2A Python SDK](#) `pip install a2a-sdk`
 -  [A2A Go SDK](#) `go get github.com/a2aproject/a2a-go`
 -  [A2A JS SDK](#) `npm install @a2a-js/sdk`
 -  [A2A Java SDK](#) using maven
 -  [A2A .NET SDK](#) using [NuGet](#) `dotnet add package A2A`
 -  [A2A Rust SDK](#) `cargo add a2a-lf`
-  Use our [samples](#) to see A2A in action

Contributing

We welcome community contributions to enhance and evolve the A2A protocol!

- **Questions & Discussions:** Join our [GitHub Discussions](#).
- **Issues & Feedback:** Report issues or suggest improvements via [GitHub Issues](#).
- **Contribution Guide:** See our [CONTRIBUTING.md](#) for details on how to contribute.
- **Private Feedback:** Use this [Google Form](#).
- **Partner Program:** Google Cloud customers can join our partner program via this [form](#).

What's next

Protocol Enhancements

- **Agent Discovery:**

- Formalize inclusion of authorization schemes and optional credentials directly within the AgentCard .